

CSA15- CLOUD COMPUTING AND BIG DATA ANALYTICS

EXPERIMENT_5

AIM

To develop a simple Cloud-Based Mark Sheet Processing System using a Cloud Service Provider to demonstrate SaaS, with options to enter student details, subjects and marks, and to calculate the total, average, percentage, rank and overall performance of each student.

PROCEDURE

1. Create a web-based Mark Sheet Processing System using a cloud Software as a Service (SaaS) platform.
2. Add fields like Name of the Student, Reg. No., Subjects, Marks, Total, Average, Percentage, Rank and Overall Performance.
3. Enter the name, register number, department and semester of the student.
4. Enter the subjects and the marks obtained in each subject.
5. Calculate the total marks, average and percentage automatically from the marks entered.
6. Determine the rank of the student and the overall performance (Distinction, First Class, Second Class or Fail).
7. Submit the form to save the mark sheet.
8. Display the processed mark sheets in the Mark Sheets report.
9. Deploy (publish) the application using the cloud service.
10. Access the application through a web browser to demonstrate SaaS.

IMPLEMENTATION

The screenshot displays the 'Mark Sheet System' web application. The interface includes a sidebar with 'Enter Marks' and 'Mark Sheets' options. The main form is titled 'Enter Marks' and contains the following sections:

- Student Details:** Fields for Name of the Student (Surendar M G), Reg. No. (1374), Department (B.Tech CSE), and Semester (Semester 5).
- Subjects and Marks (out of 100):** A table with 5 rows for subjects and their marks.

Subject	Marks
Subject 1: Cloud Computing	88
Subject 2: Big Data Analytics	92
Subject 3: Data Structures	85
Subject 4: Database Management Systems	90
Subject 5: Operating Systems	80
- Result (auto-calculated):** Fields for Total Marks (435 / 500), Average (87.00), Percentage (87.00 %), Rank (1), and Overall Performance (Distinction).

At the bottom of the form are buttons for 'Process Result', 'Submit', and 'Reset'.

Fig. 1: Mark sheet entry form with auto-calculated result

Mark Sheet System [Development]

MS

Mark Sheet System

Enter Marks

Mark Sheets

S

SURENDAR

View as: -MySelf- Trial expires in 15 days Help

Mark Sheets

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Name of the Student	Reg. No.	Subjects	Total	Average	Percentage	Rank	Overall Performance	Semester
☐ B.Tech CSE								
Surendar M G	1374	5	435	87.00	87.00 %	1	Distinction	Semester 5
Somesh D	1380	5	410	82.00	82.00 %	2	Distinction	Semester 5
Hari Pavi	6969	5	372	74.40	74.40 %	3	First Class	Semester 5
Priya R	2045	5	301	60.20	60.20 %	4	First Class	Semester 5

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Fig. 2: Mark Sheets report showing average, percentage, rank and overall performance

RESULT

The Cloud-Based Mark Sheet Processing System was successfully developed and deployed using a cloud platform, demonstrating the SaaS model.